

Kisan Mitra AI - Smart Crop Recommendation Report

Farmer Email:	divya.munnuru@gmail.com	Season:	Rabi
State:	Punjab	District:	Faridkot
Soil Type:	Black Soil	Irrigation:	Medium
Water Source:	Medium	Farm Size:	4 Acres
Previous Crop:	Wheat	Date:	28 June 2026

Seasonal Suitability Analysis

Overall Suitability Score: 88%

Agricultural analysis for Faridkot, Punjab during Rabi (Winter/October-March) season. The region's Black Soil is excellent for cultivation. With Medium irrigation availability and Medium water source, the farm is well-positioned for moderate crops.

General Advice: For Black Soil in Faridkot, ensure proper soil testing before sowing. Maintain balanced irrigation schedules. Consider crop rotation with legumes to replenish soil nitrogen.

Critical Alerts

- Monitor local weather advisories for Punjab during Rabi season for unexpected rainfall or temperature changes.
- Ensure seed treatment with fungicide before sowing to protect against soil-borne diseases common in Black Soil.
- Previous crop was Wheat. Practice crop rotation to break pest and disease cycles.

Recommended Crops

Mustard (Sarson) (Suitability Score: 90% | Water: Low | Demand: High | Growing Period: 110-140 days)

Why Recommended:	India's primary oilseed Rabi crop. Excellent for semi-arid regions with cool winters. The oil has strong domestic demand. Well-suited for Faridkot, Punjab region with Black Soil.
Water Management:	Mustard is drought-tolerant but 1-2 irrigations at flowering and pod-filling significantly boost yields. Ideal for limited water availability. Your farm has Medium irrigation and Medium water source availability.
Fertilizer Plan:	Apply DAP (40 kg/acre) as basal and Urea (25 kg/acre) at first irrigation. Apply Sulphur (20 kg/acre) for improved oil content.
Expected Yield:	6-10 quintals/acre

Chickpea (Chana/Gram) (Suitability Score: 88% | Water: Low | Demand: High | Growing Period: 90-120 days)

Why Recommended:	India's most important pulse crop. Thrives in black and medium soils with residual moisture. High protein content and strong MSP support. Well-suited for Faridkot, Punjab region with Black Soil.
Water Management:	Chickpea is very drought-tolerant and often grown rain-fed. One protective irrigation at pod-filling is beneficial. Excess water causes wilt diseases. Your farm has Medium irrigation and Medium water source availability.
Fertilizer Plan:	Rhizobium + PSB seed treatment. Apply DAP (40 kg/acre) as basal. Minimal nitrogen due to nitrogen fixation. Apply Zinc Sulphate (10 kg/acre) in deficient soils.
Expected Yield:	8-12 quintals/acre

Barley (Jau) (Suitability Score: 82% | Water: Low | Demand: Medium | Growing Period: 100-120 days)

Why Recommended:	Hardy Rabi cereal that grows where wheat fails. Dual purpose—grain for food and straw for fodder. Growing demand from the malt industry. Well-suited for Faridkot, Punjab region with Black Soil.
Water Management:	Barley is the most drought-tolerant cereal. Can produce decent yields with just 1-2 irrigations. Tolerates saline and alkaline soils. Your farm has Medium irrigation and Medium water source availability.
Fertilizer Plan:	Apply DAP (30 kg/acre) as basal and Urea (20 kg/acre) at first irrigation.
Expected Yield:	12-16 quintals/acre

■ High Risk / Unsuitable Crops

Paddy (Rice): Rice requires standing water and high temperatures (25-35°C) during growth. Rabi season's low winter temperatures cause cold injury, poor tillering, and spikelet sterility.

Cotton: Cotton needs long warm days and a frost-free growing season of 180+ days. Rabi season's declining temperatures and short days result in poor boll formation and fiber quality.